

MOHIT CHAMOLI

Graphic Era Hill University, Dehradun, Uttarakhand

 8869862542  Email  LinkedIn  GitHub  Portfolio  LeetCode

Education

Graphic Era Hill University, Dehradun
B.Tech in Computer Science & Engineering

Aug 2024 – Expected June 2028
CGPA: 8.55/10.0

Experience

PixelPulse

Feb 2026 – Present

Chief Technology Officer (CTO)

- Leading the engineering team delivering client-facing web and mobile applications, overseeing architecture decisions, code quality standards, and project delivery timelines.
- Defining technology stack selection and system architecture across multiple concurrent projects spanning React.js, Node.js, Flutter, and cloud-based deployments.
- Establishing engineering workflows including Git-based version control practices, code review processes, and CI/CD deployment pipelines.

Projects

Low-Level Systems Projects — C, Memory Management, Garbage Collection

- Implemented a custom memory allocator in C with explicit free-list management, block splitting, coalescing, and boundary tag metadata for efficient dynamic memory allocation.
- Built a mark-and-sweep garbage collector supporting root set scanning, recursive heap traversal, and automatic reclamation of unreachable objects.

BingeBuddy — React.js, Node.js, Socket.IO, WebSockets

- Developed a real-time synchronized watch-party platform with concurrent room-based sessions and distributed playback state management across multiple clients.
- Implemented event-driven architecture using Socket.IO for low-latency bidirectional communication, handling concurrent events and synchronization conflicts.

CypherFusion — Python, AES, ECC, Cryptography

- Engineered a hybrid encryption system combining AES symmetric encryption with ECC-based asymmetric key exchange for end-to-end data security.
- Implemented ECDH key agreement protocol enabling secure session key derivation without transmitting private keys.

AfwahTracker — Python, NLP, Machine Learning, scikit-learn

- Built a misinformation detection engine using NLP pipelines for text preprocessing, feature extraction, and supervised classification of news articles.
- Applied TF-IDF vectorization with machine learning classifiers for fake news detection, evaluating performance with precision, recall, and F1 metrics.

Obscura — Flutter, Dart, WireGuard, Networking

- Developed a cross-platform open-source VPN application integrating WireGuard protocol for encrypted network tunnel management and secure traffic routing.
- Implemented low-level networking interfaces bridging WireGuard's cryptographic tunnel with Flutter UI for real-time connection state monitoring.

GymQuest — Java, OOP, Design Patterns, Java Swing

- Developed a gamified fitness tracking application applying MVC architecture, Observer, and Strategy design patterns with persistent state management.

Technical Skills

- Languages:** C, C++, Java, Python, JavaScript, SQL, Dart
- Systems Programming:** Custom Memory Allocator (C), Mark-and-Sweep Garbage Collector
- Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Socket.IO
- Frontend:** React.js, HTML/CSS, Tailwind CSS, Vite
- AI/ML:** NLP, Machine Learning, spaCy, Sentence Transformers, Gemini API, scikit-learn
- Databases:** MongoDB, MySQL, Firebase Firestore
- Coursework:** Data Structures & Algorithms, Compiler Design, DBMS, Operating Systems, OOP

Achievements

- Achieved a LeetCode contest rating of 1,655 (top 17% globally) with 200+ DSA problems solved across arrays, trees, graphs, and dynamic programming.
- Maintained 1,500+ active contributions on GitHub with consistent open-source development across multiple repositories.